

ProblemSet2

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```
#### Standard Procedure & Preparation
## Clear working environment
rm(list=ls())

## Set working directory
setwd("C:/Users/onno/Desktop/TRINITY/PhD/RStudio")
```

Question 1

```
## Import our data into R, so create a matrix! we'll call it obs for observations
obs <- matrix(c(14, 7,      # Not Stopped
               6, 7,      # Bribe requested
               7, 1),     # Stopped/given warning
             nrow = 2,
             byrow = FALSE)

## make our table a bit more beautiful and easier to handle
rownames(obs) <- c("Upper Class", "Lower Class")
colnames(obs) <- c("NotStopped", "Bribe", "Stopped")
obs
```

```
##           NotStopped Bribe Stopped
## Upper Class      14     6      7
## Lower Class       7     7      1
```

SUBTASK 1

```
## calculate row and column means for chi square calcs
row_totals <- rowSums(obs) ##27 and 15!
col_totals <- colSums(obs) ##21, 13 and 8
grand_total <- sum(obs) ##42

## now we can calculate our expected number of observations that we need for chi^2
expected <- outer(row_totals, col_totals) / grand_total
## lets have a look
expected
```

```
##           NotStopped  Bribe  Stopped
## Upper Class    13.5 8.357143 5.142857
## Lower Class     7.5 4.642857 2.857143
```

```
## calculate chi^2 by estimating components and summing them
chi_sq <- sum((obs - expected)^2 / expected)
chi_sq
```

```
## [1] 3.791168
```

```
## our chi^2 is 3.799!!
```

```
### SUBTASK 2
```

```
## First of all, let's calculate the degrees of freedom that we have
```

```
#### df=(r1)(c1)
df <- (2-1)*(3-1)
## df = 2
```

```
# p-value
p_value <- pchisq(chi_sq, df, lower.tail = F)
p_value #0.15
```

```
## [1] 0.1502306
```

```
## -> we cannot reject the H0; p = 0.15, >.05
## because 0.15 > .10 we even reject at 10% level
```

```
### SUBTASK 3
```

```
std_resid <- as.data.frame((obs - expected) / sqrt(expected))
View(std_resid)
```

```
### SUBTASK 4
```

```
# they are a cell-level view of chi^2 results, so we can see exactly
# which cells have exceedingly higher or lower observed than expected outcomes!
# This gives us a finer grained view of patterns behind the single
# test statistic (chi^2)
```

Question 2

```
## Load the data into R
```

```
url <- "https://raw.githubusercontent.com/kosukeimai/qss/master/PREDICTION/women.csv"
df <- read.csv(url, header = TRUE)
```

```
## SUBTASK 1
```

```
# H0: There is no difference in the number of new (or repaired) water sources
# between villages that are lead by men and villages that are lead by women
```

```
# H1: The number of new (or repaired) water sources in villages that are lead
# lead by men is higher (or lower!) than in villages that are lead by women
```

```
## SUBTASK 2
```

```
## lets do the regression, first fit the model.. we take $female not $reserved,
## because we want to know ALL women
regression <- lm(water ~ female, data = df)
```

```
## SUBTASK 3
```

```
# lets see it
summary(regression)
```

```
##
## Call:
## lm(formula = water ~ female, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -22.68 -14.78  -7.81   2.29  317.32
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   14.813     2.382   6.220 1.56e-09 ***
## female         7.864     3.838   2.049  0.0413 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 33.51 on 320 degrees of freedom
## Multiple R-squared:  0.01295,    Adjusted R-squared:  0.009867
## F-statistic: 4.199 on 1 and 320 DF,  p-value: 0.04126
```

```
## what can we say?
## we reject H0 ("female leadership has no effect") is rejected at the 5% level
## Also: Villages governed by female council heads have, on average, about 7.86 more
## new or repaired drinking water facilities than those governed by men
```

```
## However, I am confused about the phrasing "reservation policy" - so let's check the
## effect of $reserved
```

```
regression2 <- lm(water ~ reserved, data = df)
summary(regression2)
```

```
##
## Call:
## lm(formula = water ~ reserved, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -23.991 -14.738 -7.865 2.262 316.009
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  14.738      2.286   6.446 4.22e-10 ***
## reserved      9.252      3.948   2.344 0.0197 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 33.45 on 320 degrees of freedom
## Multiple R-squared:  0.01688,    Adjusted R-squared:  0.0138
## F-statistic: 5.493 on 1 and 320 DF,  p-value: 0.0197
```

```
## we also reject H0 at 5% level
## effect here is even bigger: villages that were reserved for female council heads
## have on average 9.25 more new or repaired water facilities
```